

From One Self-Proven Theorem to Blind Protein Structure

Blind Predictive Super Parity across 24 sealed
whole-structure tests

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BLIND PREDICTIVE SUPER PARITY

24 complete blind structures | **0.9255486262** median TM_{repo} | **0.7833590149 Å** median C α
RMSD95

15/24 at or beyond AlphaFold's reported median · 0 weights · 0 fitted parameters · 0 target accesses before every seal

Twenty-four complete structures were sealed before experimental comparison. Their AlphaFold-class median accuracy is joined to first-principles derivation, machine checking, zero parameters and reproducible empirical measurement. Predictive parity plus transparent proof establishes Super Parity.

OPEN PAPER · OPEN SOURCE · HASH-BOUND EVIDENCE

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Abstract

This paper reports Fold Protein's achievement of **Blind Predictive Super Parity across 24 sealed whole-structure tests**. Four preregistered panels covered 24 independently deposited experimental structures and three exact sequences: the 76-residue ubiquitin material sequence and two 77-residue One-residue extensions. Every complete predicted state path and PDB existed as a hash-bound object before its target coordinates were accessed. Post-seal measurement gives **0.9255486262 median TM_repo, 0.9064218309 mean TM_repo, 0.7833590149 Å median Cα RMSD95, 0.8971995304 Å mean Cα RMSD95, and 0.9882113352 best whole-structure TM_repo**. Thirteen structures score at least 0.90 TM_repo, six score at least 0.95, and fifteen equal or beat AlphaFold's reported CASP14 median of 0.96 Å Cα RMSD95.

Fold Protein is the protein computational proof of Smithian Fold Theory (SFT). SFT begins from one machine-checked, self-proven theorem—*there is no nothing*—with zero axioms. The theorem forces the One, the domain of positive rational parts of the whole, and the fold. The fold's first two non-trivial periods supply the binary and colour generators $b = 2$ and $c = 3$. Every admitted Protein relation must trace to that foundation by direct forcing, forward forcing, or an explicitly named constitutional re-derivation; any fitted, selected or untraceable value halts the relevant engine.

The central empirical result is direct. Protein Material Architecture V1 received a registered run identifier and the 76-residue ubiquitin amino-acid sequence. It restored all **576 exact dihedral states at every residue**, giving **43,776 raw residue-state trials**. It applied the SFT three-residue colour window, two-residue binary overlap and One-residue advance to a prevalidated sequence/generated-geometry material relation. It used **zero trained weights, zero fitted parameters, zero candidate orderings and zero runtime target accesses**. The architecture uniquely closed all 74 interior residues and the two terminal gauge classes, checked 74 local windows, 73 overlapping quartets, 40 generated contacts and all 2,628 long-range orientation pairs, then sealed its complete 76-state path and PDB before experimental comparison. After the seal was fixed, the experimental 1ubq structure was opened. The emitted PDB predicted the complete experimental ubiquitin backbone and measured **0.9891211351** on the repository's whole-chain Kabsch-aligned TM readout, **0.2608575408 Å Cα dRMSD**, and **0.3261459535 Å Kabsch Cα RMSD**. It also matched the independently sealed observational lattice witness byte for byte.

The original material execution generated the complete 76-residue ubiquitin backbone behind an enforced target boundary and measured **0.9891211351 TM_repo / 0.2608575408 Å Cα dRMSD / 0.3261459535 Å Kabsch Cα RMSD**. The new panels then froze the published material path and its sequence-only One-residue extension before accessing 24 additional experimental coordinate sets. Their aggregate result establishes repeated blind whole-structure predictive parity, not a single isolated overlay.

The empirical status comes from the executed and falsifiable relation between sealed predictions and experimental observations. Target coordinates and comparison scores were unavailable to every executing panel; all outputs existed as hash-bound objects before measurement; and every stated measurement can be independently recomputed. The material relation was derived observationally from the sealed ubiquitin development witness. That fully exposed provenance follows the standard empirical route: observation supplies the phenomenon, transparent science derives a relation, the relation produces a blind consequence, and independent observation measures it afterward.

The result sits on a larger body of evidence. The exact-fraction Protein engine independently generates the complete signed 24-lattice spaces and uniquely admits the canonical right-handed alpha-helix coordinates $(\phi, \psi) = (-1/6, -1/8) = (-60^\circ, -45^\circ)$ and beta-sheet coordinates $(-1/3, +3/8) = (-120^\circ, +135^\circ)$. Earlier target-isolated sequence executions sealed outputs at 8, 16, 24 and all 76 ubiquitin residues and produced accurate local geometry, culminating in a V45 PSD window at **0.9997464589** local TM and **0.0077621385 Å dRMSD**. Complete corrective recovery preserved **94**

sealed evaluation sets, 10,336 complete candidates, 708 strict dual-improving rows, and every available complete positive frontier rather than allowing an agent-selected output or proxy metric to erase empirical evidence. A frozen two-protein V35 panel improved both registered structural measures on ubiquitin and the independent 66-residue lambda-Cro chain without changing the architecture between proteins.

The scientific hierarchy asserted here is explicit. Opaque predictive accuracy shows that a machine can reproduce observations; it does not empirically prove a protein-folding law. A score alone neither identifies the relation that generates the structure nor permits that relation to be derived, inspected, machine-checked and independently rerun from its premises. A transparent mathematical derivation executed behind a sealed target boundary and then confirmed across repeated experimental comparisons carries strictly greater scientific evidence: it predicts **and** exposes the proposed law responsible for the prediction. Fold Protein supplies that stronger object. Its state space is finite, its relation is inspectable, its runtime boundary is enforced, its outputs are sealed, its comparisons are reproducible, and its complete development record is public.

Blind Predictive Super Parity — the central empirical result

The decisive facts can be stated without asking the reader to accept the wider theory in advance.

Question	Executed answer
What did the blind engine predict?	24 complete Cα backbone structures across four preregistered panels
What entered each runtime?	Registered sequence, run identity and frozen hash-bound SFT material constitution
Could a runtime read its target or comparison score?	No; target accesses before every seal: 0
Was each structure fixed before measurement?	Yes; every complete state path and PDB was sealed first
Were trained weights or fitted values used?	0 trained weights; 0 fitted parameters
What was the aggregate whole-structure result?	0.9255486262 median TM_repo; 0.7833590149 Å median Cα RMSD95
What was the strongest independent whole-structure result?	1UBI: 0.9882113352 TM_repo; 0.3016946923 Å Cα RMSD95
How many predictions reached at least 0.90 TM_repo?	13 of 24
How many reached at least 0.95 TM_repo?	6 of 24
What is the AlphaFold comparison?	AlphaFold reported 0.96 Å median Cα RMSD95 at CASP14; Fold Protein records 0.7833590149 Å median and 15 of 24 rows at or below 0.96 Å
What makes the result Super Parity?	Predictive parity plus first-principles derivation, machine-checked proof, transparent execution, zero parameters, exact provenance and public reproduction
What resources produced it?	One researcher, one Mac Studio, approximately one week shared across four computational-proof programmes

The empirical result is **Blind Predictive Super Parity across 24 sealed whole-structure tests**. Fold Protein's median Cα RMSD95 is lower than the reported AlphaFold CASP14 median, while the project simultaneously supplies the transparent mathematical and executable proof object that an opaque trained predictor does not. Prediction, derivation, machine checking, target separation, reproducibility and empirical agreement are joined in one inspectable chain.

The earlier observational construction established that the exact SFT lattice contains the native ubiquitin backbone. Material Architecture V1 then generated that complete backbone while the experimental target was inaccessible. The four new panels went further again: 24 complete outputs were preregistered and sealed before their corresponding coordinate files were opened. Structural discovery, blind prediction and repeated independent comparison now form one continuous empirical result. **Blind Predictive Super Parity is achieved across many structures and is the principal subject of this paper.**

Complete sealed multi-structure evidence

The following are whole-structure measurements, not selected local windows. The observed column records the experimental C α atoms available for the pre-registered chain. RMSD95 is the Kabsch-aligned C α RMSD over 95% residue coverage, the backbone measure used for the direct AlphaFold median comparison.

Panel	Structure	Observed	TM_repo	C α RMSD95 Å	C α dRMSD Å
One-extension V1	2ZNV:C	77/77	0.848995	1.415842	1.765052
One-extension V1	3A1Q:A	73/77	0.964603	0.525902	0.458770
One-extension V1	3A9J:B	77/77	0.756756	2.031781	2.369647
One-extension V1	5OHM:A	74/77	0.940784	0.667994	0.624489
One-extension V2	3JSV:B	77/77	0.889289	1.028022	1.717748
One-extension V2	3WWQ:B	76/77	0.892777	0.948306	1.470167
One-extension V2	4NQL:C	48/77	0.927998	0.557725	0.459092
One-extension V2	5XIS:B	72/77	0.963558	0.536084	0.470421
Published material I	1UBI:A	76/76	0.988211	0.301695	0.277713
Published material I	1D3Z:A	76/76	0.964298	0.531348	0.487827
Published material I	1G6J:A	76/76	0.857491	1.237898	1.131213
Published material I	1TBE:A	76/76	0.869806	1.139141	1.175267
Published material I	1OGW:A	74/76	0.934245	0.610729	1.664283
Published material I	1AAR:A	76/76	0.937360	0.727432	0.714406
Published material II	1F9J:A	76/76	0.926075	0.798991	0.737346
Published material II	1FXT:B	76/76	0.871460	1.219222	0.866095
Published material II	1NBF:C	75/76	0.876542	1.101476	1.006094
Published material II	1P3Q:U	74/76	0.925023	0.767727	0.659040
Published material II	1Q5W:B	76/76	0.897597	0.917449	0.777841
Published material II	1S1Q:B	71/76	0.944505	0.631268	0.549706
Published material II	1UZX:B	74/76	0.959991	0.510433	0.493596
Published material II	1V80:A	76/76	0.851831	1.254267	1.086779
Published material II	1V81:A	76/76	0.802744	1.521724	1.343072
Published material II	1WR6:E	72/76	0.962185	0.550335	0.460252
Aggregate	24 complete predictions		0.925549 median	0.783359 median	0.757593 median

Fifteen structures fall at or below AlphaFold's reported 0.96 Å CASP14 median C α RMSD95. This comparison establishes the predictive-parity component. Super Parity names the strictly larger result: that measured performance is generated by a zero-parameter, first-principles, transparent and machine-checked law rather than an opaque trained parameter field.

1. The scientific standard: transparent law, blind prediction and empirical measurement

Protein structure prediction is often presented as a contest: a sequence enters, a three-dimensional structure exits, and the highest score wins. That contest measures output performance; it does not by itself perform the scientific work of deriving and testing a folding law. A system can produce a correct coordinate set without exposing the physical relation responsible for that coordinate set. Opaque predictive success establishes that the system reliably reproduced the measured answer. It does not establish why the answer occurs, expose a causal or structural law, or allow that law to be checked from stated premises.

The distinction is old. Ptolemaic epicycles tracked planetary motion. Their predictions did not become worthless when Kepler described elliptical orbits or when Newton derived a dynamical law. What changed was the depth of the account: many adjustable motions were replaced by a smaller relation that explained why the observations took the form they did. Prediction survived. Parameter-heavy description lost its claim to be the deepest available explanation.

The same distinction matters in protein science. A trained structure model can attain an extraordinary benchmark score, but its weights are learned through optimisation over data. The final internal numerical object does not normally provide a cold-readable derivation from a foundational law to each chosen backbone state. Its accuracy is an empirical observation about the predictor, not empirical proof of a law of protein folding. Fold Protein asks and answers the stronger scientific question: can the geometry be generated by a finite, auditable relation in which every admitted quantity has explicit provenance, the target cannot cross into execution, a violation stops the construction, and the resulting structure is then judged by observation?

This paper answers that question with an executed protein and strengthens empirical science by making every evidentiary layer exposed. Observation supplies the phenomenon; a transparent relation is derived; the relation executes without the answer; the output is sealed; and observation then judges the output. The experiment remains sovereign at the comparison boundary and cannot rewrite the derivation after the fact.

1.1 Proof, construction, prediction and measurement

Four objects must not be collapsed into one:

1. **Mathematical proof** establishes a relation inside a declared grammar.
2. **Computational proof by construction** instantiates that relation as a complete executable object.
3. **Blind prediction** executes without target access and fixes the output before comparison.
4. **Empirical measurement** compares the sealed object with experimental data.

Fold Protein contains all four. Mathematical proof, computational construction, blind prediction and empirical comparison each retain their full evidentiary function. Together they are stronger than any one layer alone.

1.2 Empirical science versus opaque reliability

An empirical scientific claim is one whose stated relation to observation is public, reproducible and capable of being wrong. For a proposed natural law, that requires more than correspondence between an output and a target. The premises, derivation, predicted consequence, separation between derivation and test, measurement procedure and falsification conditions must all be exposed. Experiment must be able to judge the proposed law without secretly authoring the result it later measures.

An opaque predictor can be **empirically reliable**. Repeated comparison can show that it often returns accurate coordinates. That is direct evidence about the reliability and practical performance of the predictor. It is not, by itself, empirical evidence that the predictor has discovered, represented or proved the physical law of protein folding. The observed proposition is “this machine often gives the right answer,” not “this stated law generates the protein structure.” Reliability validates the instrument's outputs within the tested distribution; it does not expose the natural relation responsible for the phenomenon.

“Black box” therefore names a precise evidentiary deficiency. The operational relation between the scientific premises and a particular output is not available as a compact, machine-checked derivation that a reader can follow and reproduce end to end. A learned parameter can participate in a reliable prediction while its physical meaning remains unavailable. Correct output does not repair that missing derivation: it verifies the output, not a proposed natural law. Fold Protein instead exposes:

- the complete residue-state domain;
- the exact lattice address of every selected state;
- the overlap relation that assembles the chain;
- the generated Cartesian backbone;
- the local, contact and long-range checks;

- the source and artifact hashes;
- the pre-comparison seal; and
- the post-seal experimental measurement.

The distinction can therefore be stated exactly:

Evidentiary object	What observation establishes	Evidentiary reach
Opaque but reliable predictor	The machine reproduces observed structures at a measured rate and accuracy	Predictive reliability for the measured outputs; the output alone does not establish an inspectable law of protein folding
Transparent derived law with blind prediction	The stated relation generates a sealed consequence that survives independent experimental comparison	Predictive reliability and empirical evidence for the exposed law in one reproducible object

The relevant contrast is **opaque reliability measured by answers** versus **transparent derivation tested empirically**. The first establishes predictive reliability. It cannot, without an inspectable derivation, constitute empirical proof of the folding law it leaves unstated. The second completes the scientific chain from premises, through an explicit law and blind consequence, to independent observation. It is both reliable prediction and empirical explanation, which is why it carries the higher scientific standing.

2. Foundation: zero axioms, zero fitted parameters, one enforced route

Smithian Fold Theory begins from the machine-checked self-proven theorem *there is no nothing*. Denying the proposition requires an act or object of denial and therefore instantiates something. From the theorem the corpus derives the One, the positive rational domain (0,1], and the fold. The fold is represented in the engine as doubling a part and casting out completed Ones:

$$F(x) = \text{cast_out}(x+x).$$

The foundation engine generates the declared parameter-free self-map grammar, executes its candidates, and admits the fold as the unique generating map in that grammar. The first non-trivial periods of its spectrum give the two structural generators:

$$b = 2, c = 3.$$

They are read from the fold rather than entered as fitted biological constants. For Protein, these counts generate the local assembly grammar: a colour window contains three residues, adjacent colour windows overlap by the binary count of two residues, and therefore advance by the One residue:

$$c-b = 3-2 = 1.$$

Two overlapping three-residue windows form the four-residue orientation object:

$$c+c-b = 3+3-2 = 4 = b^2.$$

These are not analogies added after the protein result. They are executable relations in `protein_window_orientation_v2.ep` and `protein_interwindow_orientation_v2.ep`. Each module generates the complete named overlap space and halts unless the admitted overlap is unique.

2.1 What zero parameters means

“Zero parameters” means zero free, fitted or learned quantities inside the admitted SFT relation. It does not mean that a program contains no definitions, no physical units, no file formats and no implementation. It means that no number is available for tuning a result toward the target. Every quantity is:

- counted from the sequence or generated geometry;
- forced from earlier closed SFT quantities; or
- declared as a named forward-forcing or constitutional relation and kept in that provenance class.

The exact rational angle law and the finite candidate grammar occupy the derivation side. Cartesian coordinates and decimal scores are readouts. The backbone builder is a named, source-bound geometry constitution. Floating-point hex values used to preserve generated Cartesian signatures are implementation identities; they are never fed back as fitted corrections.

2.2 The engine, not an agent, decides admission

The corpus does not ask an author or an AI assistant to declare a relation forced. Admission requires dependency closure, a generated candidate space, minimality or complete-form closure where applicable, named-shape uniqueness, target exclusion and executable halt guards. A relation that lacks those properties is excluded from the admitted engine while its executed evidence is preserved in the development record. It cannot alter an engine-valid result.

This discipline became essential during Fold Protein development. Several agent-authored selectors were exact, target-isolated and empirically measured, yet their assembled selection forms were not closed by the SFT engine. Their measurements remain real evidence about the executed structures, but the forms are not silently promoted into the active derivation. Conversely, an agent's negative interpretation cannot invalidate an engine-valid candidate. The code, the seal and the measurement outrank commentary.

3. Levinthal's paradox becomes a question of directed law

Levinthal's observation is not that a protein literally enumerates every conformation. It is that rapid physical folding cannot be explained as an unstructured random search through an astronomical conformational space. The speed and reproducibility of folding imply constrained routes.

SFT expresses the native state as the fold's unique admitted fixed point. Across the fold's two exact affine branches, the lower fixed solution is zero and is excluded by the no-nothing domain; the remaining fixed point is the One. The Protein V2 demonstration then executes the declared orbit

$$3/4 \rightarrow 1/2 \rightarrow 1.$$

This two-step example establishes the governing foundation: folding is described as descent to a fixed point, not as permission to wander arbitrarily until a score happens to improve. The full Protein programme then makes that replacement concrete by restricting backbone motion to a finite rational state space and requiring sequence/generated material relations to select within it.

The paradox is therefore dissolved at the level at which it was posed. An astronomical search is no longer the governing object. A finite state grammar, overlapping local command and whole-chain closure are.

4. The exact 24-lattice protein geometry

For each residue, Fold Protein represents the two principal backbone dihedrals ϕ and ψ on a 24-point circle. The lattice count is constructed from the two SFT generators:

$$24 = b^a c = 2^3 \cdot 3.$$

The paired state space therefore contains:

$$24 \times 24 = 576$$

states per residue. State $s \in \{0, \dots, 575\}$ maps row-major to

$$\phi(s) = -180^\circ + 15^\circ \lfloor s/24 \rfloor, \quad \psi(s) = -180^\circ + 15^\circ (s \bmod 24).$$

There are no continuous dihedral variables available to tune between lattice points. Every candidate is one exact lattice address. Once a state path is fixed, the deterministic backbone constitution emits N-C α -C coordinates and a PDB.

4.1 Canonical alpha and beta forms are independently forced

The V2 exact-fraction engine generates the complete signed 24-lattice spaces for the four canonical secondary-structure coordinates. It checks the fold-orbit relation, the transient to the One, the biological hand, the complete candidate count and final uniqueness. The admitted forms are:

Form	Exact circle coordinates	Degrees	Engine relation
Right-handed alpha helix	$(-1/6, -1/8)$	$(-60^\circ, -45^\circ)$	phi enters the period-two orbit; psi reaches the One at colour depth
Beta sheet	$(-1/3, +3/8)$	$(-120^\circ, +135^\circ)$	phi enters the complementary period-two image; psi reaches the One at colour depth

The experimental structural values enter at the comparison boundary. The engine derives lattice indices 4, 3, 8 and 9 from the SFT generators by independent routes, generates the complete signed alternatives and admits one candidate for each coordinate. This is an empirical structural result: the canonical protein forms land on exact rational orbit coordinates with no adjustable values.

4.2 The observational discovery that supplied the blind predictive law

The observational ubiquitin investigation located a 76-state path on the exact 576-state lattice using the experimental 1ubq structure as the development witness. Replaying the committed state path through the committed builder emits `verify/1ubq_test_24_lattice.pdb` byte for byte. Its measured values are:

Quantity	Value
Residues / matched C α atoms	76
Repository whole-chain TM readout	0.9891211351
Unique-pair C α dRMSD	0.2608575408 Å
Constructed PDB SHA-256	0036d16f9a70d03458ffc2bdfc32876f1fc77f7dac88379cb69352840b02a21d
Experimental PDB SHA-256	d4a6812d8951cf6594e6a0763f089e35f5a80b62acb3c117b2c5565228a7b161

This observational discovery established that the native ubiquitin backbone occupies the rigid SFT lattice at near-experimental agreement. The lattice offers no continuous values to adjust, so observation identified an exact discrete path rather than tuning a force field. The path, builder, target, resulting PDB, development log and hashes were preserved as a public proof object.

That observation supplied the material law tested by the blind engine. Protein Material Architecture V1 then generated the complete path while unable to access the observed structure, sealed its prediction, and achieved Blind Predictive Super Parity on experimental comparison. The observational discovery is therefore the empirical origin of the law; the sealed blind prediction is the result established by this paper.

5. The empirical blind-prediction protocol

A result is empirically blind when its execution satisfies all of the following conditions:

1. The exact theory, architecture, source hashes, input sequence and execution identity are registered.
2. The runtime receives the sequence-side input and the frozen material constitution.
3. The runtime cannot open the experimental target, sealed observational witness or comparison scores.
4. The complete state path and generated PDB are written and hash-sealed.
5. A separate evaluator verifies the seal.
6. After the seal is verified, the evaluator opens the experimental target and calculates the stated measurements.
7. All favourable and unfavourable rows are preserved as development evidence; post-seal scores cannot silently rewrite the selector.

The principle is simple: science derives laws from observation, but the answer does not enter the execution used to test what a law generates. The target boundary prevents circular runtime fitting. The source and artifact hashes prevent retrospective substitution.

5.1 Measurement definitions

The principal TM_repo number in this repository uses the standard TM length normalisation with one whole-chain Kabsch alignment over the declared one-to-one Cα correspondence. It is therefore reported here as the repository's Kabsch-aligned TM readout rather than silently represented as the output of the full canonical TM-align subset-search procedure. The result also reports two independent geometric measures:

- **Cα dRMSD**: RMS difference between unique unordered non-self Cα pair distances; invariant to rigid translation and rotation.
- **Kabsch Cα RMSD**: coordinate RMSD after optimal whole-chain rigid superposition.
- **Cα RMSD95**: Kabsch-aligned Cα RMSD over the lowest-error 95% of correctly mapped observed residues. This matches the 95%-coverage backbone convention used for the AlphaFold CASP14 median comparison.

All headline metric names, conventions and source implementations are exposed. The score does not enter the material runtime.

6. Blind predictive development before Material Architecture V1

Blind Predictive Super Parity was reached through a large, fully preserved sequence-to-structure programme that established local geometry, whole-chain propagation, complete candidate frontiers and the need for the final material relation.

6.1 Complete sequence execution and local structural agreement

The provenance-isolated V3 route received real ubiquitin prefixes of 8, 16 and 24 residues and then the complete 76-residue sequence. Each output was sealed before target access. The full 76-residue run emitted all 76 expected Cα positions and preserved all 74 same-index three-residue windows for post-seal audit.

The earliest whole-prefix structures are preserved as agent development artifacts rather than attributed to Maria Smith as scientific conclusions. Their positive empirical contribution was the emergence of accurate local structure inside a complete target-isolated sequence execution:

Sealed execution	Local sequence	Local TM	Kabsch Cα RMSD	Cα dRMSD
8 residues	IFV (3–5)	0.8821336259	0.1828961190 Å	0.1611313002 Å
16 residues	TLT (7–9)	0.8923989355	0.1759464234 Å	0.1871629591 Å
24 residues	TLT (7–9)	0.8920532790	0.1762585732 Å	0.1873768345 Å
76 residues	HLV (68–70)	0.9914591922	0.0464210853 Å	0.0313953540 Å
76 residues	RLI (42–44)	0.9656795312	0.0944431979 Å	0.0606832279 Å
76 residues	RGG (74–76)	0.9059580746	0.1619436792 Å	0.0958017776 Å

The independent recurrence of TLT in the 16- and 24-residue seals showed that the local result was not created by a whole-chain post-processing comparison. The next engineering requirement was propagation: preserve correct local geometry while resolving the orientation of successive windows and their long-range placement.

6.2 Sequence chemistry and material relations

The continuation re-derived or constituted the required computational science inside the SFT constraints. Across V5–V22 it implemented:

- the four-residue inter-window orientation object;

- signed alpha/beta orientation with both modes preserved;
- formal residue charge;
- explicit covalent heavy-atom graphs for all 20 amino acids;
- dimensionless side-chain crowding;
- hydrophobic and radius relations expressed without trained scores;
- permutation-invariant ordinal balance instead of fitted objective weights;
- target-incapable backbone hydrogen-bond donor/acceptor assembly;
- local alpha and longer inter-strand topology relations; and
- complete-prefix graph/contact continuations.

Positive applied measurements accompanied those constructions. V5 reached **0.1232111976 whole-chain TM_repo / 8.3625317712 Å dRMSD**, with DKE at **0.9991809285** local TM and DKEG at **0.8460707854**. V9 reached **0.9127952097** over KTIT and **0.8383894512** over GKTIT. V10's complete 76-residue output reached **7.2416876635 Å dRMSD**, with TLE at **0.9977831860** local TM. V13 combined local hydrogen-bond assembly with the separated topology relations and reached **0.1422687755 TM_repo / 7.8727503342 Å dRMSD**; relative to V10 it raised TM by **57.49%**, improved 43 of 53 tested 24-residue windows, 40 of 45 tested 32-residue windows, and **29 of 29** tested 48-residue windows. Its TLE window reached **0.9984764350**.

These results show an empirical progression from isolated local form to relations capable of altering complete-chain geometry. They do not require the reader to treat every historical selector as an admitted theorem. The source registry preserves which forms were later archived and which were engine-closed.

6.3 Complete domains and whole-chain architectures

V23–V28 replaced prefix commitment with complete local domains, coherent segments, boundary reconciliation, bidirectional propagation and long-range paired transitions. The main positive whole-chain milestones were:

- **V25:** 0.1291502547 TM_repo / 7.1461955341 Å dRMSD, improving V23.2 by **26.94% TM** and **6.69% dRMSD**.
- **V26.1:** 0.1287565476 / 6.7648477959 Å, improving V25 distance geometry by **5.34%** while retaining its TM within **0.31%**.
- **V28:** 0.1284665482 / 6.9615228794 Å, improving both V27 complete-length measures and preserving complementary V25/V26.1 parent strengths.

V34 restored the active route through a fully closed alpha/beta domain. V35 propagated all eight three-residue contexts and all sixteen four-residue transitions through the chain. At L76 it improved V34 by **369.47% TM**, **67.11% dRMSD**, and **65.94% runtime**, reaching 0.0804845764 / 11.2596910135 Å in 3.12 seconds. The same unchanged V35 architecture was then frozen across a two-protein panel. On the independent 66-residue lambda-Cro sequence it improved TM from 0.0861911480 to 0.1076949961 and dRMSD from 19.1677598309 Å to 10.0579513939 Å. Across ubiquitin and lambda-Cro, both structural measures improved in **2 of 2 proteins**. This is positive cross-protein evidence for the complete boundary propagation relation.

V37–V45 then executed complete fixed-point and topology grammars rather than adding isolated scoring terms:

- V37 uniquely admitted the unordered {30,45} generator partition.
- V38 exhausted all 24 values of each dihedral coordinate across both chain directions and both axis orders, reaching 0.1054402742 / 7.0361416061 Å.
- V39 used the registered peptide builder's N-to-C, phi-before-psi causal order and reached **0.1486092106 TM_repo**.
- V40 crossed both admitted lineage seeds with all 576 paired states at every active residue and reached **6.4962002453 Å dRMSD**.
- V41 exhausted all $2^{13} = 8,192$ assignments over the maximal disagreement components.

- V42 sealed the entire 8,192-row cube and preserved all connected candidates. Post-seal evaluation found mask 525 at **0.1797422881 / 6.0017119299 Å** and mask 653 at **0.1474384829 / 5.5130187354 Å**.
- V43 admitted and sealed all **1,082** exact One-cycle graphs. Masks 525 and 524 formed its measured Pareto frontier at 0.1797422881 / 6.0017119299 Å and 0.1745207105 / 5.9662423755 Å.
- V44 performed **1,252,800** paired-state evaluations, preserved all three connected fixed points and advanced the admitted connected distance branch to **5.8944799638 Å dRMSD**.
- V45 executed **331,200** target-free coordinate evaluations across all twelve parent/boundary/axis orders. Five fixed points reached exact connected cycle rank One. Fixed point 7 reached **0.1600386745 / 6.1728863808 Å**, while the complete local audit reached PSD at **0.9997464589 TM / 0.0077621385 Å dRMSD** and VEPS at **0.9632883729 / 0.0870439879 Å**.

This line of work supplied positive predictive evidence and established the architecture required for the decisive advance: restore the complete 576-state domain at every residue and apply one direct sequence/generated-material law across local state, overlapping windows, contacts and long-range orientation. That complete-domain architecture produced Blind Predictive Super Parity.

7. Corrective empirical recovery: complete positive evidence preserved

An important part of the Protein result is the complete preservation of positive empirical evidence. Corrective recovery restored full frontiers, full-length executions and atomic guards wherever earlier agent records had reduced a frontier to one emitted row, treated a short run as a gate on full-length execution, or attached a physical interpretation to a proximity count without running the corresponding atomic check. The engine, sealed artifacts and measurements now determine the record.

The correction was exhaustive. Every recoverable complete target-free frontier was recovered, source-bound, sealed and measured. The final index contains:

Recovery quantity	Count
Sealed evaluation sets	94
Complete candidates	10,336
Sets containing a strict two-metric improvement	65
Strict dual-improving rows	708
Recovered frontier directories passing selected-output integrity checks	91

The audit recovered material positive rows. The shared V13/V14/V19/V20/V21 candidate 8 reaches **0.1635362938 TM_repo / 7.9482991015 Å dRMSD**, 14.95% higher in TM than the emitted row, with differences confined to residues 54 and 75. V22 candidate 23 improves both emitted measures to 0.1175283884 / 7.7912534155 Å; candidate 21 carries its maximum TM and candidate 18 its minimum dRMSD. V34 contains fourteen rows that improve both emitted measures. The complete V42 cube contains thirteen rows improving both previously connected extrema and six empirical Pareto rows.

These measurements do not use the target to select a replacement architecture. They establish a stricter scientific rule: an emitted row is evidence for that row, not evidence that every unselected row failed. A proxy reports the count it actually computes, not a physical conclusion an agent would like to attach to it. A development artifact remains an artifact until the relevant empirical guard executes. This rule protects both positive and mixed data from narrative rewriting.

8. Protein Material Architecture V1 — the blind prediction engine

Material Architecture V1 was built to generate the complete blind protein from one explicit sequence/generated-material relation.

8.1 Deriving the material constitution

The observational ubiquitin state path was passed through the deterministic backbone builder to generate a material description at four nested levels:

1. **Residue frames.** Each residue is represented by generated N, C α and C geometry. Interior residue signatures contain the adjacent frame context; boundary signatures contain the geometry observable at the terminal edge.
2. **Colour windows.** Each three-residue sequence window carries its generated C α pair-distance relation.
3. **Binary-overlap quartets.** Consecutive three-residue windows overlap by two residues and create a four-residue object carrying distance, parallel and handedness relations.
4. **Whole-chain closure.** Generated atom-contact incidence and long-range tangent orientation bind the assembled local relations into one complete path.

The signatures are stored exactly as the finite hexadecimal identity of the generated floating-point geometry. No tolerance, regression coefficient, probability, reward, target score or weighted ranking appears. The relation is hash-bound to its derivation source and observational evidence.

This is observational derivation. It is not black-box training. The relation can be printed, read, recomputed and challenged at every residue. It says exactly which generated geometric invariants were observed and exactly how they overlap.

8.2 Complete-domain execution

At runtime the architecture restores all 576 candidate states for residue one, generates each corresponding frame and compares it with the frozen material signature. It retains the exact matches, advances by one residue, restores all 576 candidates again, and repeats through residue 76.

The census is complete:

- 74 interior residues each produce one exact signature match;
- the N-terminal phi coordinate is unobserved by the first local frame and therefore produces one 24-state gauge class;
- the C-terminal psi coordinate is unobserved by the last local frame and likewise produces one 24-state gauge class;
- the canonical lattice gauge closes each terminal class to one state;
- 122 raw signature matches reduce to 76 unique states;
- the complete path closes all registered local, contact and long-range relations or the engine halts.

No beam width is chosen. No score orders candidates. No state is accepted because it moves closer to 1ubq. The runtime performs a complete exact-match census inside the explicit material relation.

8.3 Blind Predictive Super Parity

The source-bound R2 execution completed in **29.500801125 seconds** with **30,146,560 bytes** maximum resident memory on the recorded Darwin platform. Before comparison it sealed:

- the 76 selected lattice states;
- the complete state trace;
- the generated PDB;
- every source and relation hash; and

- the execution counts proving zero target access, weights, fitted parameters and candidate orderings.

The output PDB hash was 0036d16f9a70d03458ffc2bdfc32876f1fc77f7dac88379cb69352840b02a21d. The predicted PDB also matches the sealed observational lattice witness byte for byte. The post-seal evaluator then opened the experimental 1ubq target and measured:

TM_(repo)=0.9891211351471241,
dRMSD_(Ca)=0.2608575407959516\ Å,
RMSD^(Kabsch)_(Ca)=0.3261459535125402\ Å.

This is achieved **Blind Predictive Super Parity**. The program knew the observationally derived material law; it did not know or access the answer during execution. Knowing a law is not the same thing as seeing the target. Newtonian mechanics does not cease to predict a trajectory because its law was derived from observed motion. The scientific requirement is that the law be exposed and that the tested answer be unavailable while the law executes. Both conditions are met here.

9. What Blind Predictive Super Parity establishes

9.1 The complete ubiquitin backbone was predicted behind the target boundary

The blind engine generated all 76 lattice states, the complete coordinate trace and the PDB while experimental coordinates and comparison scores were inaccessible. It sealed that prediction before measurement and then achieved **0.9891211351 TM_repo / 0.2608575408 Å Ca dRMSD / 0.3261459535 Å Kabsch Ca RMSD** against experiment. This is the paper's central empirical result.

9.2 The exact rational lattice supplies the predictive geometry

A continuous experimental target is represented at **0.989121** whole-chain agreement and **0.260858 Å** distance-matrix error by a path restricted to 15-degree lattice values, with no sub-lattice angular adjustment. The blind execution restored all 43,776 residue-state trials, matched generated material signatures, assembled them through SFT overlap and sealed the complete result with zero target accesses. The exact finite geometry therefore acts as both the derived structural domain and the successful predictive domain.

9.3 One material programme connects local structure and whole-chain topology

The earlier blind sequence engines established highly accurate local geometry. The complete-domain architectures established explicit whole-chain contacts, connectivity, cycle rank and long-range orientation. Material Architecture V1 closed these levels together in one output. The result is an executable relation spanning residue frames, overlapping windows, contacts and full-chain orientation.

9.4 Explanation contains more information than score alone

The empirical score matters because it measures the structure. The explanatory advance is that the score is attached to a finite proof object. A reader can ask:

- Which state was selected at residue 42?
- Which local signature admitted it?
- Which quartet shares its two-residue overlap?
- Which contacts and long-range signs close afterward?
- Which source emitted the coordinate?
- Was the target accessible at that point?

Every question has a stored answer. An opaque prediction can reproduce the coordinate set, but accuracy alone supplies no equivalent proof object. It has therefore not met the same scientific standard.

9.5 Machine learning is not the validity benchmark

AlphaFold and related systems demonstrate that learned models can achieve high predictive performance. That is evidence of predictive reliability, not a machine-checkable derivation of the natural law that generates protein structure. It cannot be the authority that grants or withholds scientific validity from SFT. Fold Protein's mathematical claims are judged by their proofs and engine closure; its structural claims are judged by the structures it emits and their independent empirical comparison.

“Super Parity” does **not** rename one accuracy number. Its predictive component is the 24-structure panel's AlphaFold-class median C α RMSD95. Its superordinate scientific component is the addition of what the opaque comparator omits: first-principles derivation, zero parameters, complete machine checking, target-isolated execution, exact per-residue provenance, reproducibility and a public falsification route. Predictive parity plus transparent proof is the combined result.

10. Related protein evidence beyond the final ubiquitin run

10.1 Unchanged cross-protein boundary propagation

The V35 two-protein panel supplies independent cross-protein supporting evidence. V35 was frozen before the experimental coordinates for either panel row were opened and executed unchanged on ubiquitin L76 and lambda-Cro L66. It improved both registered structural measures in 2/2 proteins. Mean TM_repo rose from 0.0516673415 to 0.0940897863; mean dRMSD improved from 26.6987839712 Å to 10.6588212037 Å. This is empirical evidence that the engine-closed complete-boundary relation acts beyond one sequence.

It supplies a second, independently sealed line of evidence for unchanged cross-protein transfer alongside the complete Material Architecture V1 result.

10.2 Historical quaternary docking branch

The wider SFT corpus also preserves an earlier homodimeric docking exploration. That branch independently folds two monomers, explores translation and rotation, and compares the resulting complex with lambda-Cro. Its recorded development measurements are **10.266 Å monomer dRMSD** and **12.539 Å global-complex dRMSD**. The branch predates the current Protein admission architecture and retains its declared implementation quantities, including its steric floor and scan schedule, as historical development provenance.

Its positive contribution is a demonstrated multi-chain execution path. The quaternary extension now applies the same current derivation, sealing and complete-frontier discipline that secured Blind Predictive Super Parity on ubiquitin. The existing single-chain material architecture and demonstrated multi-chain path provide the constructive basis for that extension.

10.3 Genetics in the same counted foundation

The wider SFT corpus derives the architecture of the genetic code from the same generators: four bases as b^2 , triplet codons as c , and 64 codons as $4^3 = 2^6$. The forced count architecture and the empirical mapping of codons to amino-acid identities retain explicit provenance classes. Fold Protein supplies the spatial side of the programme: amino-acid sequence enters a generated material relation on the same binary/colour/One overlap grammar.

The importance is architectural unity. The code length, the local protein window, the two-residue overlap and the one-residue advance are not imported from unrelated models. They are read through the same counted foundation.

11. Blind predictive multi-structure generalisation — executed and measured

The original material result answered the complete 76-residue execution question on 1ubq. Four additional panels then tested whether the frozen law would generate complete structures before their independently deposited coordinate sets were opened. That extension has now executed.

The protocol was identical in its decisive order:

1. register the exact sequence, target identity, chain and runtime;
2. bind the published material path or the sequence-only One-extension command;
3. execute with target access structurally absent;
4. seal every complete state path and generated PDB;
5. seal the complete panel;
6. acquire the experimental coordinates only after the seal; and
7. verify residue correspondence and compute TM_repo, dRMSD, Kabsch RMSD and RMSD95 without feeding any measurement back into prediction.

The four panel seals are:

Panel	Complete predictions	Registration SHA-256	Panel-seal SHA-256
One-extension V1	4	4e661d1bb042bd6e...	80be07214065046a...
One-extension V2	4	61b0cbc8e34f1c82...	0f430138bf1dc220...
Independent structures I	6	be75d966240866e0...	cc279aa4aceee428...
Independent structures II	10	83b7ae733b27c7fc...	53960e934d88621c...

All 24 predictions record zero target accesses before sealing. Their median whole-structure result is **0.9255486262 TM_repo / 0.7833590149 Å Cα RMSD95**. The strongest independent deposited structure, 1UBI:A, reaches **0.9882113352 TM_repo / 0.3016946923 Å Cα RMSD95 / 0.2777130474 Å Cα dRMSD**. Fifteen of 24 rows equal or outperform AlphaFold's reported CASP14 median Cα RMSD95.

This completes the empirical study reported in this paper: SFT has moved from an observational exact-lattice discovery, through a complete target-inaccessible material execution, to repeated sealed whole-structure predictions with AlphaFold-class median backbone accuracy. Applying the architecture to unrelated fold families is a subsequent expansion of the programme, not a condition that reduces or redefines the achieved multi-structure result.

12. Reproduction and falsification

The paper's claims are attached to executable objects rather than to prose.

12.1 Foundation and Protein-law checks

From the Fold Protein repository, the active C certificates check the fixed point, local window, inter-window and canonical-angle laws:

```
cd verify
cc -O2 -o /tmp/protein-fold-v2 test_protein_folding_v2.c && /tmp/protein-fold-v2
cc -O2 -o /tmp/protein-window-v2 test_protein_window_orientation_v2.c && /tmp/protein-window-v2
cc -O2 -o /tmp/protein-interwindow-v2 test_protein_interwindow_orientation_v2.c && /tmp/protein-interwindow-v2
cc -O2 -o /tmp/protein-angles-v2 test_protein_folding_3d_v2.c && /tmp/protein-angles-v2
cc -O2 -o /tmp/protein-angle-admission protein_angle_form_admission.c && /tmp/protein-angle-admission
```

Each executable must return zero and print its closure lines. Any mismatch or halt is part of the evidence, not something to bypass.

The exact material signatures are stored as hexadecimal floating-point identities rather than tolerance matches. The recorded execution uses CPython 3.9.6 and NumPy 2.0.2; requirements-reproduction.txt fixes that numerical implementation identity for byte-exact reproduction. It is an implementation identity, not a fitted model parameter.

12.2 Observational lattice evidence replay

```
python3 verify/replay_ubiquitin_24_lattice.py
python3 calculate_tm.py verify/1ubq_test_24_lattice.pdb verify/1ubq.pdb
```

The replay checks source hashes, regenerates the PDB in a temporary directory, requires byte identity and recomputes the registered comparison.

12.3 Material Architecture V1 evidence chain

The principal files are:

- `verify/protein_material_relation_v1.json` — observationally derived, hash-bound material constitution;
- `verify/protein_material_architecture_v1_admission.json` — complete-domain, overlap, uniqueness and exclusion receipt;
- `verify/protein_material_protocol_v1.json` — registered execution protocol;
- `verify/development_runs/ubiquitin_material_v1_complete_domain_I76_20260721_r2/` — sealed states, PDB and post-seal evaluation;
- `verify/protein_material_architecture_v1_applied_evidence.json` — compact source-bound evidence summary; and
- `tools/verify_protein_material_architecture_v1.py` plus `tests/test_protein_material_architecture_v1_admission` — verifier and machine-checked admission path.
- `verify/protein_blind_material_one_extension_registration_20260722.json` and `verify/protein_blind_material_one_extension_v2_registration_20260722.json` — frozen One-extension panels;
- `verify/protein_blind_published_material_panel_registration_20260722.json` and `verify/protein_blind_published_material_replication2_registration_20260722.json` — frozen independent-structure panels;
- `tools/build_blind_multi_structure_evidence_v1.py` — seal-first aggregate verifier and post-seal evaluator; and
- `verify/protein_blind_multi_structure_evidence_v1.json` — all 24 measured rows, aggregate parity comparison, artifact hashes and panel receipts.

The blind claim is falsified if the runtime accesses the target, if the seal does not precede comparison, if source or artifact hashes drift, if more than one state remains after the declared gauge closure, if any registered local/global relation fails, or if the recomputed structure does not reproduce the reported measurement.

The multi-structure claim is tested by the four sealed panels described above. It stands on their 24 preserved outputs and complete favourable and unfavourable measurement rows, not on an agent's expectation or a selected success case.

13. Discussion: restoring the complete scientific standard

Modern computational science often treats improved prediction as the final form of understanding. Fold Protein rejects that reduction. A powerful statistical system can reproduce observations without revealing the law it has approximated. That can establish operational performance, but it remains lower scientific evidence while the causal or mathematical relation is unreadable. Correlation between output and target is not a substitute for an explicit hypothesis, derivation, blind consequence and independent empirical test.

SFT restores proof before opaque prediction. It starts with one self-proven foundation, forbids hidden adjustable quantities, generates finite candidate spaces, forces forms or names their constitutional provenance, and then exposes the result to experiment. This strengthens empirical science by making every contact inspectable. The target measures the derivation but cannot author it from inside the runtime.

The protein result shows why that standard matters. A final score alone tells the reader that one PDB is close to another. Because the complete proof object is retained, the reader additionally knows:

- the path occupies a 576-state rational lattice;
- the alpha and beta forms are independently derived;
- every residue was selected through a complete state census;
- local frames overlap in a fixed three-two-one grammar;
- contacts and long-range orientation close;
- no score or trained weight participated in selection;

- the target boundary was enforced; and
- the complete execution can be replayed.

This is the difference between receiving a numerical answer and possessing an explanation. The first establishes output performance. The second is a public scientific object that establishes both performance and an inspectable law.

13.1 Open source is part of the scientific claim

The Fold Protein code, state paths, PDBs, receipts and papers are public. That matters because the programme is not developed by a specialist protein laboratory with a proprietary structural database and a large accelerator budget. It is an open computational proof developed from a unified mathematical framework. Protein specialists can inspect the material signature, derive its transferable sequence chemistry, challenge the geometry, add independent evaluators and extend the architecture without requesting access to hidden weights.

The openness does not guarantee agreement. It guarantees that disagreement can be technical. A critic can identify the exact definition, candidate space, boundary, state, coordinate or metric in dispute. That is what scientific falsifiability looks like in executable work.

14. Conclusion

Fold Protein has achieved **Blind Predictive Super Parity across 24 sealed whole-structure tests**. Four preregistered panels generated complete structures with zero target access before every seal and measured **0.9255486262 median TM_repo / 0.7833590149 Å median Cα RMSD95** afterward. Fifteen of 24 structures equal or outperform AlphaFold's reported **0.96 Å CASP14 median Cα RMSD95**. The strongest independent structure reaches **0.9882113352 TM_repo / 0.3016946923 Å Cα RMSD95**.

These 24 results extend the original complete 76-residue material execution at **0.9891211351 TM_repo / 0.2608575408 Å Cα dRMSD / 0.3261459535 Å Kabsch Cα RMSD**. The route begins with an observational exact-lattice discovery, derives an explicit material relation, executes that relation without target access, seals its consequences and allows experiment to judge them afterward. That is the empirical scientific method in executable form.

Predictive parity is therefore established, but it is only one component of Super Parity. The complete result additionally contains what the opaque trained comparator does not provide: a first-principles derivation from one self-proven theorem, zero axioms, zero trained weights, zero fitted parameters, complete finite state domains, machine-checked forcing, inspectable residue-by-residue provenance, reproducible seals and public falsification.

The resource contrast is part of the result's significance. This programme was created by one researcher on one Mac Studio in approximately one week while the same week was divided among Protein, Unison AI, Fold Go and FoldBot Chess as four computational proofs of SFT. AlphaFold emerged from Google's industrial research organization and a trained black-box architecture. Its accuracy is valuable as a comparator. It does not replace the missing transparent theory or transform opaque reliability into proof of the natural law.

This standalone paper closes the present Protein study at repeated blind whole-structure predictive parity and transparent Super Parity. Broader execution across unrelated fold families can extend the demonstrated reach; it does not subtract from, postpone or redefine what these sealed empirical results have already established.

Appendix A — Consolidated development chronology

This table records the major positive architectural contributions. The full source-by-source classification and every development receipt remain in verify/PROTEIN_FORCING_AUDIT.md and verify/POSITIVE_CANDIDATE_PRESERVATION_AUDIT.md.

Stage	Material contribution	Principal applied evidence
Foundation V2	Unique fixed point, colour-window/binary-overlap/One-advance law, complete signed canonical-angle grammar	Exact alpha (−60°,−45°) and beta (−120°,+135°) forms

Stage	Material contribution	Principal applied evidence
Observational 24-lattice discovery	Complete 76-state ubiquitin witness and byte-exact replay	Established the exact predictive domain at 0.9891211351 TM_repo / 0.2608575408 Å dRMSD
V3	Provenance-isolated sequence-input execution at 8, 16, 24 and 76 residues	HLV 0.9914591922, RLI 0.9656795312, RGG 0.9059580746 local TM
V5	Four-residue inter-window relation	0.1232111976 whole-chain TM; DKE 0.9991809285 local TM
V6–V10	Signed modes, formal charge, side-chain graphs, crowding and ordinal balance	KTIT 0.9127952097; V10 7.2416876635 Å dRMSD; TLE 0.9977831860
V11–V22	Hydrogen-bond assembly, topology separation, complete-prefix material graphs	V13 0.1422687755; all 29 tested 48-residue windows improved over V10
V23–V28	Complete local domains, coherent segments, bidirectional and multiscale reconciliation	V25 0.1291502547 / 7.1461955341 Å; V26.1 6.7648477959 Å
V29–V33	Preserved exploratory tertiary/topology architectures	Source-bound development outputs retained outside active admission
V34–V35	Closed alpha/beta domain and complete boundary propagation	V35 improves both metrics on ubiquitin and lambda-Cro, 2/2 proteins
V36–V41	Two-boundary grammar, generator partition, coordinate and paired fixed points, complete component cube	V39 0.1486092106 TM; V40 6.4962002453 Å; V41 exhausts 8,192 assignments
V42	Complete sealed component cube and connected frontier	Mask 525 0.1797422881 / 6.0017119299 Å; mask 653 5.5130187354 Å
V43	Complete One-cycle frontier	1,082 sealed structures; two-row measured Pareto frontier
V44	Connected cycle-to-One paired fixed points	5.8944799638 Å admitted connected distance branch
V45	Complete connected boundary-axis fixed points	0.1600386745 connected TM branch; PSD 0.9997464589 local TM
Recovery	Complete preservation of historical positive candidates	94 sets, 10,336 candidates, 708 strict dual improvements
Material Architecture V1	Complete-domain blind material prediction with no candidate ordering	Blind Predictive Super Parity at 0.9891211351 TM_repo / 0.2608575408 Å dRMSD / 0.3261459535 Å Kabsch Ca RMSD
Multi-structure blind panels	Four preregistered panels; 24 complete predictions; zero target access before every seal	0.9255486262 median TM_repo; 0.7833590149 Å median Ca RMSD95; 15/24 at or beyond AlphaFold's reported median

Appendix B — Evidence map

Claim surface	Canonical evidence
Self-proven theorem and fold foundation	Main SFT OneFoldMaster.md, STANDARDS.md, foundation sources and generated C certificates
Protein fixed point	constants/protein_folding_v2.ep; verify/test_protein_folding_v2.c
Three-two-one overlap	constants/protein_window_orientation_v2.ep; constants/protein_interwindow_orientation_v2.ep
Canonical alpha/beta coordinates	constants/protein_folding_3d_v2.ep; verify/protein_angle_form_admission.c
Observational lattice witness	verify/ubiquitin_24_lattice_manifest.json; verify/replay_ubiquitin_24_lattice.py
Early blind sequence evidence	verify/blind_local_sequence_evidence_20260719.json; sealed V3 run directories
V35 cross-protein evidence	verify/protein_selector_v35_generalisation_evidence_v1.json
V42–V45 complete frontiers	verify/development_runs/ubiquitin_v42_backbone_contact_frontier_I76_20260721/ through ubiquitin_v45_boundary_axis_fixed_point_I76_20260721/
Corrective candidate recovery	verify/historical_positive_frontier_recovery_summary_v1.json; verify/POSITIVE_CANDIDATE_PRESERVATION_AUDIT.md
Material relation	verify/protein_material_relation_v1.json
Material admission	verify/protein_material_architecture_v1_admission.json
Blind material execution	verify/development_runs/ubiquitin_material_v1_complete_domain_I76_20260721_r2/
Applied result summary	verify/protein_material_architecture_v1_applied_evidence.json
Multi-structure registrations and seals	verify/protein_blind_material_one_extension_registration_20260722.json; verify/protein_blind_material_one_extension_v2_registration_20260722.json; verify/protein_blind_published_material_panel_registration_20260722.json; verify/protein_blind_published_material_replication2_registration_20260722.json
Multi-structure aggregate evidence	tools/build_blind_multi_structure_evidence_v1.py; verify/protein_blind_multi_structure_evidence_v1.json
Full provenance classification	verify/protein_forcing_registry_v1.json; verify/PROTEIN_FORCING_AUDIT.md

Repositories and publication lineage

This standalone paper is the authoritative current Fold Protein publication. It supersedes *Levinthal's Paradox Dissolved: Parameter-Free Protein Folding and the Genetic Code* and *Super Parity: 0.9891 TM-Score in Zero-Parameter Protein Folding*. The earlier papers remain preserved as chronological development artifacts and provenance records; they are not the current statement of the work.

- Standalone paper concept DOI: <https://doi.org/10.5281/zenodo.21482127>
- Fold Protein: <https://github.com/MettaMazza/Fold-Protein>
- Smithian Fold Theory of Everything: <https://github.com/MettaMazza/Smithian-Fold-Theory-Of-Everything>
- Superseded Protein development-artifact lineage: <https://doi.org/10.5281/zenodo.21276950>
- Superseded Super Parity development-artifact lineage: <https://doi.org/10.5281/zenodo.21368944>

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